

Name: Sagi Sahasra Srihasa
Roll no: 2520030008

Step 1:

```
1  /* Name: Sagi Sahasra Srihasa
2     Roll no: 2520030008
3     */
4
5     -- Step 1: Create Database
6 • CREATE DATABASE joins_operations_db;
7 • USE joins_operations_db;
8     -- Step 2: Create Tables
9 • CREATE TABLE class(
10     id INT,
11     name VARCHAR(30)
12 );
13
14 • CREATE TABLE class_info(
15     id INT,
16     address VARCHAR(30)
17 );
```

✓	56	19:01:53	CREATE DATABASE joins_operations_db	1 row(s) affected
✓	57	19:01:56	USE joins_operations_db	0 row(s) affected
✓	58	19:01:58	CREATE TABLE class(id INT, name VARCHAR(30))	0 row(s) affected
✓	59	19:02:01	CREATE TABLE class_info(id INT, address VARCHAR(30))	0 row(s) affected

Step 2:

```
18     -- Step 3: Insert Data
19 • INSERT INTO class VALUES
20     (1,'Sahasra'),
21     (2,'Tejaswini'),
22     (3,'Angela'),
23     (4,'Bhanu Priya'),
24     (5,'Nikshitha');
25
26 • INSERT INTO class_info VALUES
27     (1,'DELHI'),
28     (2,'MUMBAI'),
29     (3,'CHENNAI'),
30     (7,'NOIDA'),
31     (8,'PANIPAT');
```

✓	61	19:07:44	INSERT INTO class VALUES (1,'Sahasra'), (2,'Tejaswini'), (3,'Angela'), (4,'Bhanu Priya'), (5,'Nikshitha')	5 row(s) affected
✓	62	19:08:34	INSERT INTO class_info VALUES (1,'DELHI'), (2,'MUMBAI'), (3,'CHENNAI'), (7,'NOIDA'), (8,'PANIPAT')	5 row(s) affected

Step 3:

```
32      -- Step 4: CROSS JOIN
33 •    SELECT *
34      FROM class
35      CROSS JOIN class_info;
36      -- Step 5: INNER JOIN
```

Result Grid				
Filter Rows:				
Export:  Wrap				
id	name	id	address	
5	Nikshitha	1	DELHI	
1	Sahasra	8	PANIPAT	
1	Sahasra	7	NOIDA	
1	Sahasra	3	CHENNAI	
1	Sahasra	2	MUMBAI	
1	Sahasra	1	DELHI	
2	Tejaswini	8	PANIPAT	
2	Tejaswini	7	NOIDA	
2	Tejaswini	3	CHENNAI	
2	Tejaswini	2	MUMBAI	
2	Tejaswini	1	DELHI	
3	Angela	8	PANIPAT	
3	Angela	7	NOIDA	

Step 4;

```
36      -- Step 5: INNER JOIN
37 •    SELECT *
38      FROM class
39      INNER JOIN class_info
40      ON class.id = class_info.id;
41
```

Result Grid				
Filter Rows:				
Export:  Wrap Cell Content: 				
id	name	id	address	
1	Harshika	1	DELHI	
2	Pranitha	2	MUMBAI	
3	Angela	3	CHENNAI	
1	Sahasra	1	DELHI	
2	Tejaswini	2	MUMBAI	
3	Angela	3	CHENNAI	

Step 5:

```
43  -- Step 6: INNER JOIN Selected Columns
44  • SELECT class.name,
45         class_info.address
46  FROM class
47  INNER JOIN class_info
48  ON class.id = class_info.id;
49
50
51  -- Step 7: NATURAL JOIN
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	name	address
▶	Harshika	DELHI
	Pranitha	MUMBAI
	Angela	CHENNAI
	Sahasra	DELHI
	Tejaswini	MUMBAI
	Angela	CHENNAI

Step 6:

```
51  -- Step 7: NATURAL JOIN
52  • SELECT *
53  FROM class
54  NATURAL JOIN class_info;
55
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	id	name	address
▶	1	Harshika	DELHI
	2	Pranitha	MUMBAI
	3	Angela	CHENNAI
	1	Sahasra	DELHI
	2	Tejaswini	MUMBAI
	3	Angela	CHENNAI

Step 7:

```
58  -- Step 8: LEFT OUTER JOIN
59  • SELECT *
60  FROM class
61  LEFT OUTER JOIN class_info
62  ON class.id = class_info.id;
63
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	id	name	id	address
▶	1	Harshika	1	DELHI
	2	Pranitha	2	MUMBAI
	3	Angela	3	CHENNAI
	4	Bhanu Priya	NULL	NULL
	5	Nikshitha	NULL	NULL
	1	Sahasra	1	DELHI
	2	Tejaswini	2	MUMBAI
	3	Angela	3	CHENNAI
	4	Bhanu Priya	NULL	NULL
	5	Nikshitha	NULL	NULL

Step 8:

```
66 -- Step 9: LEFT JOIN with NULL
67 • SELECT *
68 FROM class
69 LEFT JOIN class_info
70 ON class.id = class_info.id
71 WHERE class_info.id IS NULL;
72
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [FA](#)

	id	name	id	address
▶	4	Bhanu Priya	NULL	NULL
	5	Nikshitha	NULL	NULL
	4	Bhanu Priya	NULL	NULL
	5	Nikshitha	NULL	NULL

Step 9:

```
74 -- Step 10: RIGHT OUTER JOIN
75 • SELECT *
76 FROM class
77 RIGHT OUTER JOIN class_info
78 ON class.id = class_info.id;
79
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [FA](#)

	id	name	id	address
▶	1	Sahasra	1	DELHI
	1	Harshika	1	DELHI
	2	Tejaswini	2	MUMBAI
	2	Pranitha	2	MUMBAI
	3	Angela	3	CHENNAI
	3	Angela	3	CHENNAI
	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Step 10:

```
--
82 -- Step 11: RIGHT JOIN with NULL
83 • SELECT *
84 FROM class
85 RIGHT JOIN class_info
86 ON class.id = class_info.id
87 WHERE class.id IS NULL;
88
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [FA](#)

	id	name	id	address
▶	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Step 11:

```
92 • SELECT *
93 FROM class
94 LEFT JOIN class_info
95 ON class.id = class_info.id
96
97 UNION
98
99 SELECT *
100 FROM class
101 RIGHT JOIN class_info
102 ON class.id = class_info.id;
103 -- Step 13: Fill OUTER JOIN Unmatched Records
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [FA](#)

	id	name	id	address
▶	1	Harshika	1	DELHI
	2	Pranitha	2	MUMBAI
	3	Angela	3	CHENNAI
	4	Bhanu Priya	NULL	NULL
	5	Nikshitha	NULL	NULL
	1	Sahasra	1	DELHI
	2	Tejaswini	2	MUMBAI
	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Step 12:

```
104 • SELECT *
105 FROM class
106 LEFT JOIN class_info
107 ON class.id = class_info.id
108 WHERE class_info.id IS NULL
109
110 UNION
111
112 SELECT *
113 FROM class
114 RIGHT JOIN class_info
115 ON class.id = class_info.id
116 WHERE class.id IS NULL;
117 -- Step 14: Create Set Operation Tables
118 • CREATE TABLE first_table(
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [FA](#)

	id	name	id	address
▶	4	Bhanu Priya	NULL	NULL
	5	Nikshitha	NULL	NULL
	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Step 13:

```

117 -- Step 14: Create Set Operation Tables
118 • CREATE TABLE first_table(
119     id INT,
120     name VARCHAR(30)
121 );
122
123 • CREATE TABLE second_table(
124     id INT,
125     name VARCHAR(30)
126 );
127 -- Step 15: Insert Set Operation Data
128 • INSERT INTO first_table VALUES
129     (1,'Sahasra'),
130     (2,'Manu');
131
132 • INSERT INTO second_table VALUES
133     (2,'Manu'),
134     (3,'Chinni');

```

✓ 76 19:15:04 CREATE TABLE first_table(id INT, name VARCHAR(30))
 ✓ 77 19:15:06 CREATE TABLE second_table(id INT, name VARCHAR(30))
 ✓ 78 19:15:15 INSERT INTO first_table VALUES (1,'Sahasra'), (2,'Manu')
 ✓ 79 19:15:17 INSERT INTO second_table VALUES (2,'Manu'), (3,'Chinni')

Step 14:

```

135 -- Step 16: UNION
136 • SELECT *
137 FROM first_table
138
139 UNION
140
141 SELECT *
142 FROM second_table;

```

Result Grid		
	id	name
▶	1	Sahasra
	2	Manu
	3	Chinni

Step 15:

```

143  -- Step 17: UNION Names
144  •  SELECT name
145      FROM first_table
146
147      UNION
148
149      SELECT name
150      FROM second_table;
151  -- Step 18: UNION ALL

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	name
▶	Sahasra
	Manu
	Chinni

Step 16:

```

151  -- Step 18: UNION ALL
152  •  SELECT *
153      FROM first_table
154
155      UNION ALL
156
157      SELECT *
158      FROM second_table;
159
160
161

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	id	name
▶	1	Sahasra
	2	Manu
	2	Manu
	3	Chinni

Step 17:

```

162  -- Step 19: Count UNION ALL Records
163  •  SELECT COUNT(*)
164      FROM (
165          SELECT * FROM first_table
166
167          UNION ALL
168
169          SELECT * FROM second_table
170      ) AS A;
171

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	COUNT(*)
▶	4

Step 18:

```

175      -- Step 20: INTERSECT
176 •    SELECT *
177      FROM first_table
178
179      INTERSECT
180 ✖    SELECT *
181      FROM second_table;

```

Result Grid			Filter Rows:		Export:		Wrap Cell Content:	
	id	name						
▶	2	Manu						

Step 19:

```

185      -- Step 21: INTERSECT Names
186 •    SELECT name
187      FROM first_table
188
189      INTERSECT
190
191 ✖    SELECT name
192      FROM second_table;
193      -- Step 22: MINUS
194 •    SELECT f.*
195      FROM first_table f
196      WHERE NOT EXISTS (

```

Result Grid			Filter Rows:		Export:		Wrap Cell Content:	
	name							
▶	Manu							

Step 20:

```

193      -- Step 22: MINUS
194 •    SELECT f.*
195      FROM first_table f
196      WHERE NOT EXISTS (
197          SELECT 1
198          FROM second_table s
199          WHERE s.id = f.id
200          AND s.name = f.name
201      );

```

Result Grid			Filter Rows:		Export:		Wrap Cell Content:	
	id	name						
▶	1	Sahasra						

Step 21:

```
202      -- Step 23: MINUS Names
203 •    SELECT f.name
204      FROM first_table f
205     WHERE NOT EXISTS (
206         SELECT 1
207         FROM second_table s
208         WHERE s.name = f.name
209     );
```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	name
▶	Sahaera